LLM FunSearch

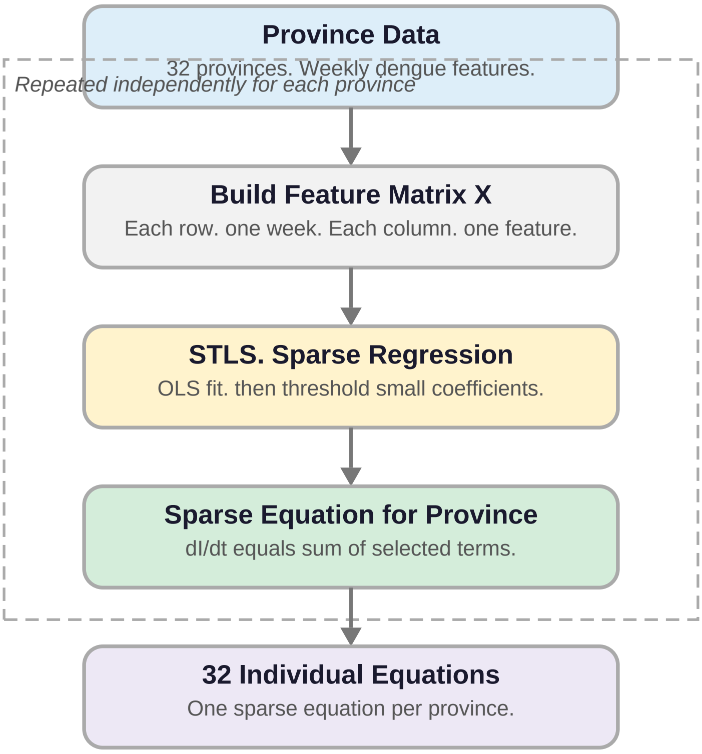
Architecture Evolution

Three generations of equation discovery for dengue incidence in Dominican Republic provinces.

Architecture 1. SINDy M1
Architecture 2. Two island FunSearch
Architecture 3. GA SINDy with Simulated Annealing

Architecture 1. SINDy M1

Sparse identification of nonlinear dynamics. One equation fitted per province independently.

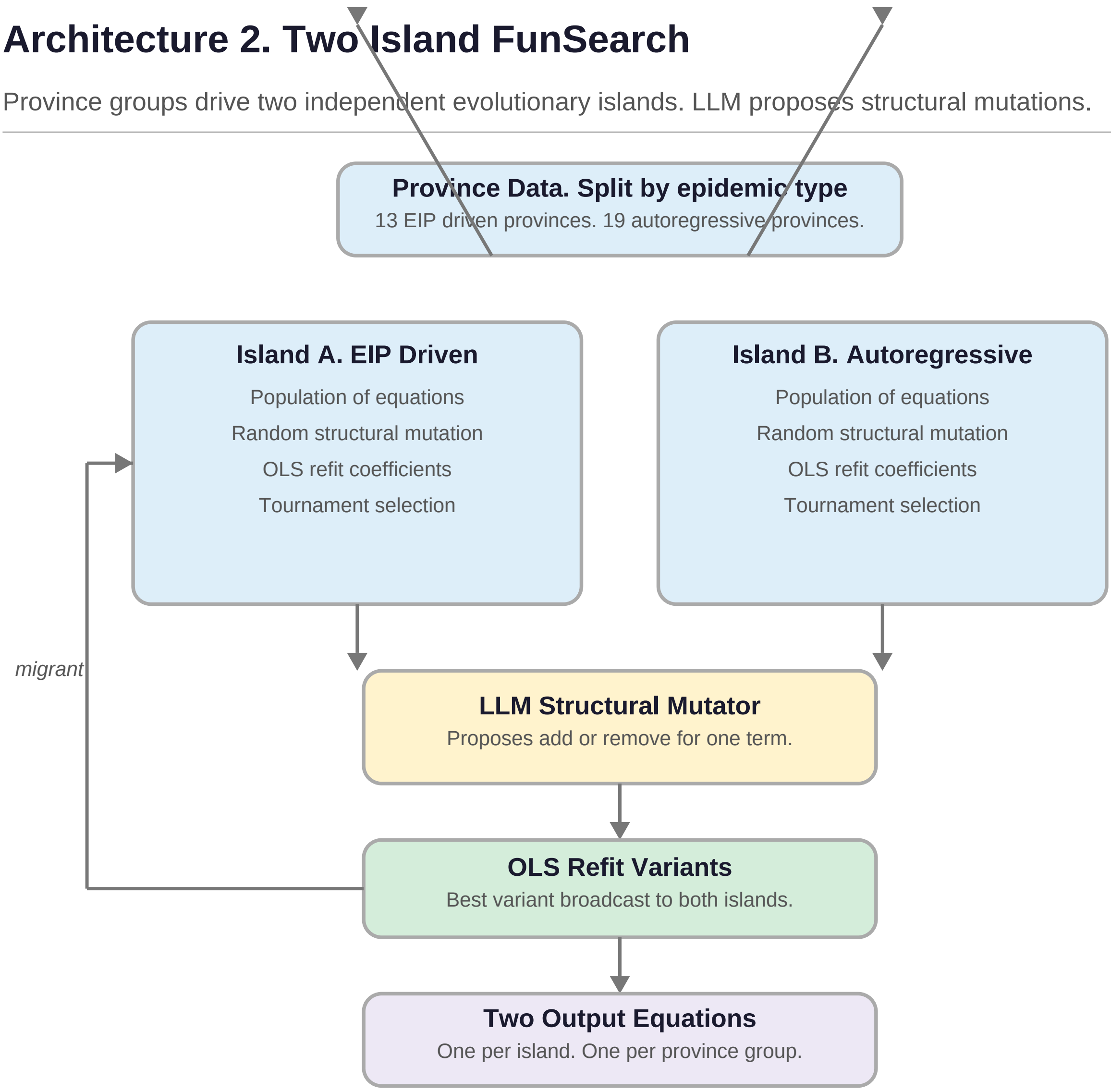


Key facts

- No evolutionary search. Coefficients and structure found in one step.
- STLS iterates OLS and thresholding until a sparse equation stabilises.
- Each province gets its own equation. No sharing of information across provinces.
- Baseline for comparison. All downstream architectures aim to beat this.

Architecture 2. Two Island FunSearch

Province groups drive two independent evolutionary islands. LLM proposes structural mutations.

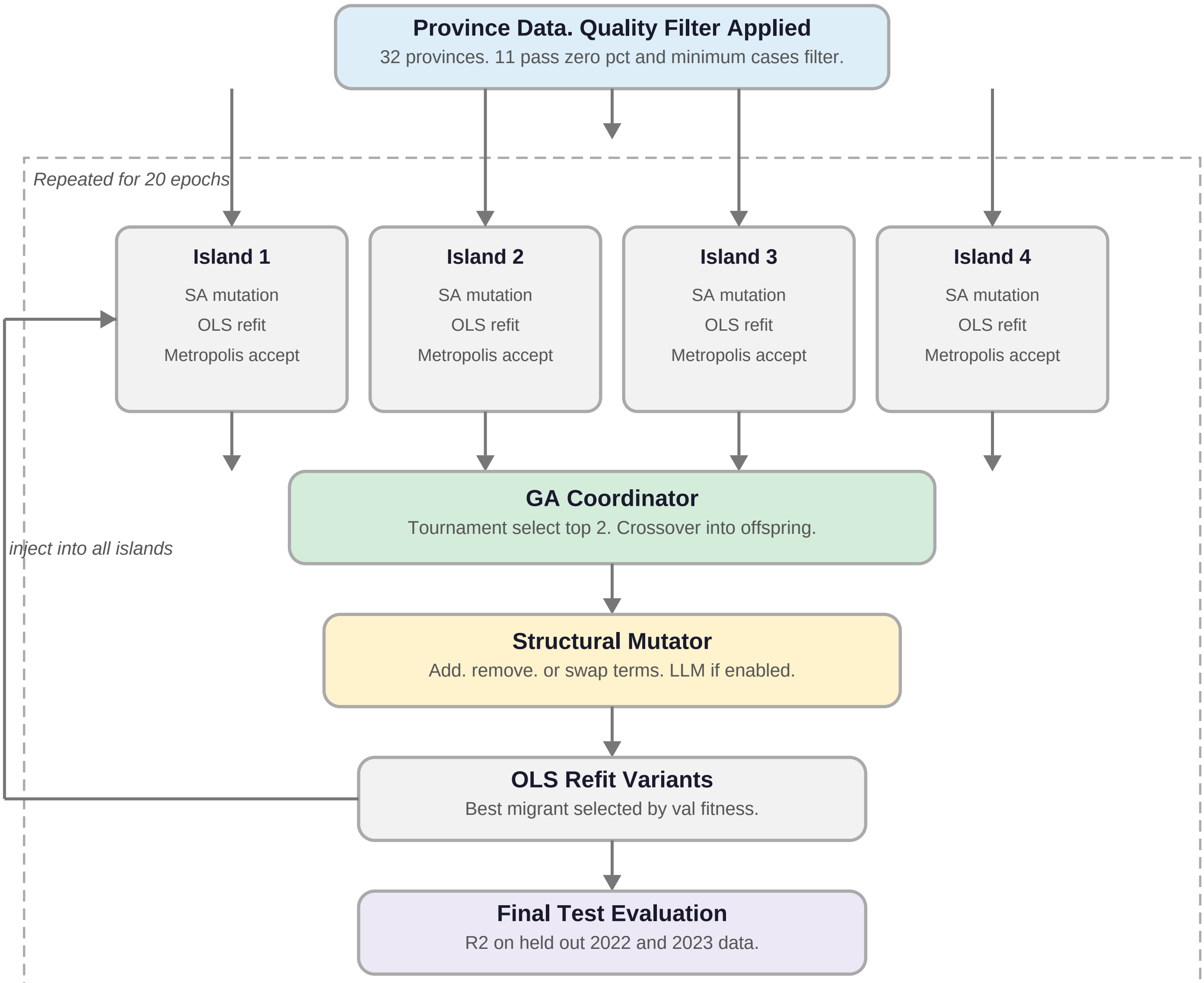


Key facts

- Islands represent province groups. not parallel searches.
- The two island split was later found to be a search artifact. not a biological signal.
- Permutation test p value was 0.19. no statistically significant grouping.
- EIP equation outperformed autoregressive equation across all temperature bands.

Architecture 3. GA SINDy with Simulated Annealing

4 parallel islands using SA search. GA coordinator drives crossover and migration.



Key facts

- Islands are parallel searches. not province groups. All islands share all 11 provinces.
- SA replaces random trial and error. Metropolis criterion allows escape from local optima.
- GA coordinator combines the best discoveries from different island trajectories.
- One shared equation output. generalises across all surviving provinces.